1. A particle of mass m moves under the influence of the attractive central force

$$\mathbf{F}(r) = -\frac{k}{r^2}\hat{\mathbf{r}},$$

where k is a positive constant. Take the potential energy to be zero at infinite separation:

$$U(\infty) = 0.$$

(a.) Explain why this force is a central force.

It is directed radially around a force center.

(b.) Using:

$$\mathbf{F}(r) = -\frac{\mathrm{d}U}{\mathrm{d}r}\hat{\mathbf{r}},$$

determine the potential-energy function $U(r)$. Clearly show the integral and apply the condition $U(\infty) = 0$.

$$U(r) = -\int F(r)\,dr = -\int_{\infty}^{r} -\frac{K}{r^2}\,dr = \int_{\infty}^{r} \frac{K}{r^2}\,dr = K\int_{\infty}^{r} r^{-2}\,dr$$

$$U(r) = \left[-\frac{K}{r^{+1}}\right]_{\infty}^{r} = \left[-\frac{K}{r} + \frac{K}{\infty}\right]$$

$$U(r) = -\frac{K}{r}$$

(c.) The particle is released from rest at a distance r_0 from the force center. Write the particle's initial total mechanical energy.

$$E_i = T_i + U_i = 0 + \left(-\frac{K}{r_0}\right)$$

$$E_i = -\frac{K}{r_0}$$

(d.) Use conservation of energy to find the particle's radial speed $|\dot{r}|$ when it reaches a smaller distance r, where $r < r_0$.

$$E_f = \frac{1}{2}m\dot{r}^2 + \left(-\frac{K}{r}\right)$$

$$\frac{1}{2}m\dot{r}^2 - \frac{K}{r} = -\frac{K}{r_0}, \quad |\dot{r}| = \sqrt{\frac{2K}{m}\left(\frac{1}{r} - \frac{1}{r_0}\right)}$$

2. Two particles, each of mass m, move along the x-axis and interact through a spring. The potential energy of the two-particle system is

$$U(r) = \frac{1}{2}k(r-\ell)^2,$$

where $r = x_2 - x_1$ is the distance between the particles, ℓ is the spring's equilibrium length, and k is a positive constant.

Initially, the particles are at rest and separated by a distance

$$r_0 = \ell + d,$$

where $d > 0$. The particles are then released.

(a.) Find the initial total energy of the system.

$E_i = T_i + U_i = 0 + \frac{1}{2}k(\ell + d - \ell)^2 = \frac{1}{2}kd^2$

(b.) Explain why the total momentum of the two-particle system is conserved. What is its value?

The force is totally internal so momentum is conserved

$P = m\dot{x}_{1i} + m\dot{x}_{2i} = 0$

(c.) Use momentum conservation to prove $\dot{x}_2 = -\dot{x}_1$.

$P_f = P_i \quad m\dot{x}_1 + m\dot{x}_2 = 0$

$m\dot{x}_1 = -m\dot{x}_2$

$\dot{x}_1 = -\dot{x}_2$

(d.) Use conservation of energy to find $\dot{x}_1$ and $\dot{x}_2$ when $r = \ell$. Assume particle 1 is initially to the left of particle 2.

$E_i = T_i + U_i = 0 + \frac{1}{2}kd^2$

$E_f = T_f + U_f = \frac{1}{2}m\dot{x}_1^2 + \frac{1}{2}m\dot{x}_2^2 + \frac{1}{2}k(\ell-\ell)^2 \quad \dot{x}_1 = -\dot{x}_2$

$E_f = \frac{1}{2}m\dot{x}_1^2 + \frac{1}{2}m\dot{x}_1^2 = m\dot{x}_1^2$

$m\dot{x}_1^2 = \frac{1}{2}kd^2$

$\dot{x}_1 = \sqrt{\frac{k}{2m}d^2} = d\sqrt{\frac{k}{2m}}$

$\dot{x}_2 = -d\sqrt{\frac{k}{2m}}$